# Online Causal-Graph-Conditioned State Space Models for Non-Stationary Multivariate Time Series Forecasting

Bachelor Thesis Proposal

University of Europe for applied science

Department of Science

Author: Aadip Thapaliya

Date: July 2026

TABLE OF CONTENTS

# 1. Introduction

Multivariate time series forecasting plays a great role in the field of energy, finance, climate and other industries. But most of the current classical machine learning models like *ARIMA , SARIMA , GARCH and Deep Learning Models like Transformers* and similar assume the causal relationships between variables stay fixed over time but in Real world they often don't: policy changes, regime shifts, or external shocks alter how variables influence each other, causing static models to degrade their efficiency.

This thesis proposes OCG SSM (Online Causal Graph Conditioned State Space Models), which learns the causal graph and forecasts together, in real time, instead of treating causal discovery as a separate offline step. As causal relationships shift, the model updates immediately without needing retraining.

Three key advantages of OCG SSM are online adaptation which enable the model to track the causal drift live , uncertainty aware which helps the model to quantify confidence in both the casual graph and forecast and at last scalable thanks to the ssm backbone it can handle long horizons efficiency.

# 2. Problem statement

We have a multivariate time series: D variables, observed over time. We want to predict what happens next, H steps ahead.

The catch: the relationships between variables aren't fixed. The way variables influence each other today might not hold tomorrow. Policies change, systems shift, and the underlying causal graph, the map of who affects whom, evolves along with the data.

Most existing approaches don't handle this well. Static causal discovery methods learn one causal graph from historical data and stick with it, even after the real structure has changed. Offline causal forecasting methods like *CDT*, *split causal discovery* and forecasting into separate stages. When the data shifts, they need to be recomputed from scratch. Anomaly focused methods, like TECamba and CGT, are built to flag unusual causal in patterns, not to forecast accurately as normal behaviour keeps changing.

State space models, like S Mamba and C Mamba, are efficient at modelling sequences, but they don't explicitly account for causal structure at all. So the real question this thesis asks is:

Can we build a forecasting model that tracks how causal relationships change over time, uses that evolving structure to guide its predictions, and still runs efficiently enough for real time use?

# 3. Research objective

## 3.1 Primary objective

To develop an online causal graph conditioned state space model that enables accurate multivariate time series forecasting under non-stationary causal dynamics by integrating online causal graph discovery with adaptive state space forecasting in a unified framework.

## 3.2 Research question

1. How can we efficiently update the estimated causal graph in an online streaming setting without recomputing from scratch, and what theoretical guarantees can we provide regarding the tracking accuracy of time-varying causal structures?
2. What is the optimal architectural design for conditioning a state space model on a dynamically evolving causal graph, and how should the conditioning be structured to preserve the linear complexity advantages of SSMs?
3. How can we detect and adapt to regime changes in the causal structure, and what mechanisms enable smooth versus abrupt transitions in the conditioning graph?
4. How can we jointly model uncertainty in both the causal graph estimates and the resulting forecasts, and what calibration properties can be achieved?
5. Under what conditions does explicit causal-graph conditioning provide measurable forecasting advantages over implicit correlation-based methods, and how robust is the approach to graph estimation errors?

# 4. Related work

## 4.1 Casual discovery in Time Series

Causal discovery for time series has evolved from classic methods like PC and FCI to based approaches like NOTEARS and neural methods like SYNOTEARS .The latest CDT uses attention to learn casual graphs but only once . If the real structure changes it requires full retraining.TECamba and CGT handle nonstationary data, but they’re designed for anomaly detection and root cause analysis, not forecasting, and neither updates the graph continuously.

## 4.2 State Space Model for time series forecasting

State space models. Mamba made sequence modelling fast and linear in cost. Variants like S Mamba, C Mamba, TSMamba, and Chimera adapted this to time series and perform well. But none of them explicitly model causal structure. They pick up correlations, not causation, and can't track how relationships shift.

## 4.3 Non stationery time series forecasting

Nonstationary forecasting. Some methods normalize away distribution shifts (RevIN, SAN). Others decompose trend and seasonality (Autoformer, FEDformer). Others adapt across time domains (AdaRNN, DTAF).

DTAF is the current best, but like the rest, it treats variables as plain features, not causally linked ones.

# 5. Methodology

## 5.1 Online Casual Graph Discovery

The foundation of our approach is an online causal graph discovery module that maintains and updates a directed acyclic graph (DAG) which is the representation of causal relationships between variables. Unlike batch methods that compute the graph from scratch, our module performs incremental updates:

Streaming Causal Score: We define a causal score function S(G; X_{t-w:t}) that measures how well a candidate graph G explains the recent window of observations. This score decomposes into two terms:

- A structural likelihood term measuring how well the graph predicts observed transitions.
- A complexity regularization term penalizing graph density to maintain sparsity.

Online DAG Update: At each time step, we perform a constrained gradient update on the graph adjacency matrix: G_{t+1} = Proj_DAG(G_t - eta_t * grad S(G_t)). The projection ensures acyclicity is maintained. We employ a first-order approximation that runs in O(D^2) per step, making it suitable for online deployment.

Regime Change Detection: We maintain a running average of the causal score and its variance. Significant deviations (exceeding k standard deviations) trigger a structural adaptation flag, indicating potential regime change in the causal dynamics.

## 5.2 Casual Graph Conditioned state space architecture

The core idea of the thesis is a state space model that adjusts its behaviour based on the current estimate of the casual graph. It has three parts that work together.

**Graph encoder**

We encode the causal adjacency matrix A_t using a graph neural network: h_G = GNN(A_t, X_t). This gives us an embedding that captures the causal structure at that point in time.

**Causal State Transition**

We write this as h_{t+1} = f(h_t, x_t; theta(A_t)), where theta(A_t) means the state space matrices (A, B, C, D) depend on the graph. In particular, the state matrix A is split into two parts: A = A_base + A_causal * M(A_t). Here M(A_t) is a mask built from the causal adjacency, so the model's transitions stay consistent with the causal structure instead of ignoring it.

**Selective Scan with Causal Bias**

The selection parameters (Delta_t, B_t, C_t) also depend on the graph embedding:

Delta_t = g_Delta(x_t, h_G)
B_t = g_B(x_t, h_G)
C_t = g_C(x_t, h_G)

This way, what the model pays attention to at each step isn't just driven by the input, it's also guided by which variables are the actual causal drivers in the current regime.

## 5.3 Adaptive Non Stationery Handling

To handle non-stationarity, we adapt at three timescales.

**Fast timescale**

We update the causal graph a bit at every step and adjust the SSM to match. This lets the model react quickly to changing causal relationships.

**Slow timescale (meta-learning)**

We remember past causal regimes. When a regime change happens, the model either jumps to a similar past regime or adapts gradually to a new one, using a memory of stored prototype graphs.

**Uncertainty-aware adaptation**

Instead of committing to one graph, we keep a distribution over possible graphs, $P(G_t \mid X_{1:t})$, and forecast by averaging over it. This captures both graph uncertainty and natural randomness in the process.

## 6. Expected Contributions

This thesis is expected to contribute to time series forecasting and causal machine learning in three ways:

**Theoretical**

- Define the problem of causal-graph-conditioned online forecasting and when it helps.
- Analyze how accurately the model tracks the true causal graph over time.
- Prove causal conditioning keeps the model's linear computational complexity.

**Methodological**

- OCG-SSM: the first state space model combining online causal discovery with causal conditioning.
- An efficient online graph update algorithm (O(D^2) per step) for streaming data.
- A multi-timescale adaptation scheme for handling different types of non-stationarity.

# 7. Evaluation plan

## 7.1 Datasets

- Synthetic: known causal graphs with controlled non-stationarity (e.g., SVAR with regime changes)
- Real-world: partial causal knowledge (electricity, traffic, climate)
- Stress tests: datasets with injected causal regime changes

## 7.2 Baselines

- Static SSMs: S-Mamba, C-Mamba, TSMamba
- Offline causal: CDT, NOTEARS-time
- Non-stationary: DTAF, AdaRNN, RevIN+Transformer
- Causal anomaly detection (adapted): TECamba, CGT

## 7.3 Metrics

- Forecasting: MSE, MAE, CRPS
- Causal graph accuracy: SHD, F1, precision, AUROC
- Adaptation: regret over time, recovery time, detection accuracy
- Efficiency: inference latency per step, memory footprint, FLOPs
- Novel: Causal Forecasting Consistency, measuring how well forecast gains track causal graph accuracy

# 8. Thesis Timeline

**August ( Literature Review & Setup)**

Finalize comparisons (CDT, TECamba, Trio, CGT, INCAMA), set up infrastructure, prepare datasets.

**September ( Online Causal Discovery)**

Build streaming causal score function and online DAG updates. Validate on synthetic ground-truth data.

**October (SSM Architecture)**

Build Graph Encoder, causal-conditioned state transitions, and selective scan with causal bias.

**November ( Non-Stationarity & Uncertainty)**

Add regime-change detection, multi-timescale adaptation, and uncertainty quantification

**December (Evaluation & Baselines)**

Run full benchmarks against S-Mamba, C-Mamba, TSMamba, CDT, DTAF, AdaRNN, TECamba, CGT. Compute SHD, CRPS, MSE/MAE, regret, recovery time.

**January (Analysis & Writing)**

Run ablations, analyze failure modes, finalize thesis and defense prep.

# 9. References


Gu, A., & Dao, T. (2023). Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*.

Zheng, X., Aragam, B., Ravikumar, P., & Xing, E. P. (2018). DAGs with NO TEARS: Continuous optimization for structure learning. *Advances in Neural Information Processing Systems, 31*.

Wang, Z., Kong, F., Feng, S., Wang, M., Yang, X., Zhao, H., Wang, D., & Zhang, Y. (2024). Is Mamba effective for time series forecasting? (S-Mamba). *Neurocomputing*. *arXiv preprint arXiv:2403.11144*.

Zeng, C., et al. (2024). CMamba: Channel correlation enhanced state space models for multivariate time series forecasting. *arXiv preprint arXiv:2406.05316*.

Ma, H., Chen, Y., Zhao, W., Yang, J., Ji, Y., Xu, X., Liu, X., Jing, H., Liu, S., & Yang, G. (2024). A Mamba foundation model for time series forecasting (TSMamba). *arXiv preprint arXiv:2411.02941*.

Du, Y., Wang, J., Feng, W., Pan, S., Qin, T., Xu, R., & Wang, C. (2021). AdaRNN: Adaptive learning and forecasting of time series. *Proceedings of the 30th ACM International Conference on Information & Knowledge Management (CIKM '21)*, 402–411.

Kim, T., Kim, J., Tae, Y., Park, C., Choi, J., & Choo, J. (2022). Reversible instance normalization for accurate time-series forecasting against distribution shift (RevIN). *International Conference on Learning Representations (ICLR)*.

Wu, H., Xu, J., Wang, J., & Long, M. (2021). Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. *Advances in Neural Information Processing Systems, 34*, 22419–22430.

Zhou, T., Ma, Z., Wen, Q., Wang, X., Sun, L., & Jin, R. (2022). FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting. *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 27268–27286.

(CDT) Causal Decomposition Transformer: causal-aligned transformer with dynamic structure learning for multivariate time series forecasting. *arXiv preprint arXiv:2505.16308*.

Khosravinia, P., Gama, J., & Veloso, B. (2026). Causally-constrained probabilistic forecasting for time-series anomaly detection (CGT). *arXiv preprint arXiv:2604.17998*.

DTAF: temporal and frequency stabilization for non-stationary time series forecasting. *arXiv preprint arXiv:2511.08229*.

# 10. Appendix

**Notation**

D- number of variables
H- forecast horizon
X_t- observation at time t
A_t - causal adjacency matrix at time t
G_t - causal graph at time t
h_G - graph embedding
h_t -SSM hidden state at time t
theta(A_t)- state space parameters conditioned on the causal graph
A_base- A_causal base and causal components of the state matrix
M(A_t) - mask derived from the causal adjacency
Delta_t, B_t, C_t - selective scan parameters
P(G_t | X_{1:t}) - posterior distribution over the causal graph
S(G; X_{t-w:t}) - causal score function
Eta_t - learning rate for the online update
SHD - Structural Hamming Distance
CRPS - Continuous Ranked Probability Score